

Algorytmy i struktury danych, Teleinformatyka, I rok

Raport z laboratorium nr: A5

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1. W pole poniżej wklej najważniejszy (według Ciebie) fragment kodu źródłowego z zajęć (maksymalnie 15 linii).

```
for i in range(n):
    if not visited[i]:
        distance = euclidean_distance(cities[current_city_index], cities[i])
        if distance < min_distance:
            min_distance = distance
            next_city_index = i

visited[next_city_index] = True
total_distance += min_distance
path.append(cities[next_city_index][0])
current_city_index = next_city_index
```

Dla mnie ten fragment jest najważniejszy, ponieważ dzięki niemu zrozumiałem jak działa algorytm zachłanny podczas szukania najlepszej ścieżki. Teraz wiem, że wyszukuje on sąsiada o najkrótszej możliwej ścieżce i do niego przechodzi. Zauważyłem również, że ten algorytm zwraca uwagę tylko i wyłącznie na konkretne miejsce i ścieżki wychodzące z niego a nie bierze pod uwagę dostępnych opcji z całej sieci.

Uzasadnij swój wybór.

2. Podsumuj wyniki uzyskane podczas wykonywania ćwiczenia. Co ciekawego zauważyłeś? Czego się nauczyłeś? Jeśli instrukcja zawierała pytania, odpowiedz na nie. Do sprawozdania możesz dodać wykresy, tabele jeśli jest taka potrzeba.

1) Algorytm, który wykorzystałem to 2-opt. Algorytm ten iteracyjnie usuwa dwie krawędzie z trasy i ponownie łączy powstałe ścieżki w sposób prowadzący do zmniejszenia całkowitej długości trasy. Zaczyna on od początkowej trasy i wybiera krawędzie, które mają zostać usunięte. Następnie ponownie łączy dwa miasta i ocenia nową trasę.

Jeśli jest krótsza to zmiana zostanie zastosowana.

2)

For order in file TSP.txt algorithm

```
['1', '16.218', '64.432', '2', '79.428', '37.861', '3',  
'31.122', '81.158', '4', '52.853', '53.283', '5',  
'16.565', '35.073', '6', '60.198', '93.9', '7', '26.297',  
'87.594', '8', '65.408', '55.016', '9', '68.921',  
'62.248', '10', '74.815', '58.704', '11', '45.054',  
'20.774', '12', '8.3821', '30.125', '13', '22.898',  
'47.092', '14', '91.334', '23.049', '15', '15.238',  
'84.431', '16', '82.582', '19.476', '17', '53.834',  
'22.592', '18', '99.613', '17.071', '19', '7.8176',  
'22.766', '20', '44.268', '43.57', '21', '10.665',  
'31.11', '22', '96.19', '92.338', '23', '0.46342',  
'43.021', '24', '77.491', '18.482', '25', '81.73',  
'90.488', '26', '86.869', '97.975', '27', '8.4436',  
'43.887', '28', '39.978', '11.112', '29', '25.987',  
'25.806', '30', '80.007', '40.872', '31', '43.141',  
'59.49', '32', '91.065', '26.221', '33', '18.185',  
'60.284', '34', '26.38', '71.122', '35', '14.554',  
'22.175', '36', '13.607', '11.742', '37', '86.929',  
'29.668', '38', '57.97', '31.878', '39', '54.986',  
'42.417', '40', '14.495', '50.786', '41', '85.303',  
'8.5516', '42', '62.206', '26.248', '43', '35.095',  
'80.101', '44', '51.325', '2.922', '45', '40.181',  
'92.885', '46', '7.5967', '73.033', '47', '23.992',  
'48.861', '48', '12.332', '57.853', '49', '18.391',  
'23.728', '50', '23.995', '45.885', '51', '41.727',  
'96.309', '52', '4.9654', '54.681', '53', '90.272',  
'52.114', '54', '94.479', '23.159', '55', '49.086',  
'48.89', '56', '48.925', '62.406', '57', '33.772',  
'67.914', '58', '90.005', '39.552', '59', '36.925',  
'36.744', '60', '11.12', '98.798', '61', '78.025',  
'3.7739', '62', '38.974', '88.517', '63', '24.169',  
'91.329', '64', '40.391', '79.618', '65', '9.6455',  
'9.8712', '66', '13.197', '26.187', '67', '94.205',  
'33.536', '68', '95.613', '67.973', '69', '57.521',  
'13.655', '70', '5.978', '72.123', '71', '23.478',  
'10.676', '72', '35.316', '65.376', '73', '82.119',  
'49.417', '74', '1.5403', '77.905', '75', '4.3024',  
'71.504', '76', '16.899', '90.372', '77', '64.912',  
'89.092', '78', '73.172', '33.416', '79', '64.775',  
'69.875', '80', '45.092', '19.781', '81', '54.701',  
'3.0541', '82', '29.632', '74.407', '83', '74.469',  
'50.002', '84', '18.896', '47.992', '85', '68.678',  
'90.472', '86', '18.351', '60.987', '87', '36.848',  
'61.767', '88', '62.562', '85.944', '89', '78.023',  
'80.549', '90', '8.1126', '57.672', '91', '92.939',  
'18.292', '92', '77.571', '23.993', '93', '48.679',  
'88.651', '94', '43.586', '2.8674', '95', '44.678',  
'48.99', '96', '30.635', '16.793', '97', '50.851',
```

```

'97.868', '98', '51.077', '71.269', '99', '81.763',
'50.047', '100', '79.483', '47.109', '']
Distance between 7 and 8 is 50.90183105743839
Distance between 8 and 9 is 8.040086628886533
Distance between 9 and 10 is 6.877439349060076
Distance between 10 and 11 is 48.212052652837755
Distance between 11 and 12 is 37.845335916199765
Distance between 12 and 13 is 22.32913885061401
Distance between 13 and 14 is 72.53655592182469
Distance between 14 and 15 is 97.76682024081586
Distance between 15 and 16 is 93.56477093970786
Distance between 16 and 17 is 28.916378749767397
Distance between 17 and 18 is 46.11071764785275
Distance between 18 and 19 is 91.97188965200182
Distance between 19 and 20 is 41.969489824871594
Distance between 20 and 21 is 35.83871103987977
Distance between 21 and 22 is 105.18266781651813
Distance between 22 and 23 is 107.68353916684016
Distance between 23 and 24 is 80.84188643058992
Distance between 24 and 25 is 72.13066724355183
Distance between 25 and 26 is 9.080996090738056
Distance between 26 and 27 is 95.26833214221817
Distance between 27 and 28 is 45.48207348351656
Distance between 28 and 29 is 20.289448415370984
Distance between 29 and 30 is 56.081590170037096
Distance between 30 and 31 is 41.30050701867957
Distance between 31 and 32 is 58.33983319311087
Distance between 32 and 33 is 80.44738882648709
Distance between 33 and 34 is 13.587504149033407
Distance between 34 and 35 is 50.35536798594565
Distance between 35 and 36 is 10.475891274731712
Distance between 36 and 37 is 75.48150210482035
Distance between 37 and 38 is 29.04320541882387
Distance between 38 and 39 is 10.953299822427944
Distance between 39 and 40 is 41.346840774114774
Distance between 40 and 41 is 82.44705820925329
Distance between 41 and 42 is 29.096975477873983
Distance between 42 and 43 is 60.292221140044255
Distance between 43 and 44 is 78.86704597612365
Distance between 44 and 45 is 90.65059351708626
Distance between 45 and 46 is 38.15545191044132
Distance between 46 and 47 is 29.207729218307954
Distance between 47 and 48 is 14.724525934643875
Distance between 48 and 49 is 34.658723375219694
Distance between 49 and 50 is 22.854703345263527
Distance between 50 and 51 is 53.45094573531885
Distance between 51 and 52 is 55.536525085388625
Distance between 52 and 53 is 85.34521364763229
Distance between 53 and 54 is 29.259030640128866
Distance between 54 and 55 is 52.17862407154869
Distance between 55 and 56 is 13.516958866549826
Distance between 56 and 57 is 16.123010667986296

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Distance between 57 and 58 is 62.98057901448668
 Distance between 58 and 59 is 53.15422150685682
 Distance between 59 and 60 is 67.20563176550013
 Distance between 60 and 61 is 116.2147090768204
 Distance between 61 and 62 is 93.30795035049265
 Distance between 62 and 63 is 15.069683772395488
 Distance between 63 and 64 is 20.007518711724348
 Distance between 64 and 65 is 76.22271236639378
 Distance between 65 and 66 is 16.697858601928576
 Distance between 66 and 67 is 81.34066550625215
 Distance between 67 and 68 is 34.465771904891376
 Distance between 68 and 69 is 66.34339144180073
 Distance between 69 and 70 is 77.94349153713863
 Distance between 70 and 71 is 63.890404670811094
 Distance between 71 and 72 is 55.966313475161115
 Distance between 72 and 73 is 49.44906965757799
 Distance between 73 and 74 is 85.46632692288817
 Distance between 74 and 75 is 6.971513279769318
 Distance between 75 and 76 is 22.686466352431353
 Distance between 76 and 77 is 48.03005901516258
 Distance between 77 and 78 is 56.28538510128539
 Distance between 78 and 79 is 37.41347738449341
 Distance between 79 and 80 is 53.82220104194922
 Distance between 80 and 81 is 19.290465640051305
 Distance between 81 and 82 is 75.628639412659
 Distance between 82 and 83 is 51.048610108405484
 Distance between 83 and 84 is 55.60933760619703
 Distance between 84 and 85 is 65.44308919970082
 Distance between 85 and 86 is 58.32814204138513
 Distance between 86 and 87 is 18.513438605510323
 Distance between 87 and 88 is 35.29500141663122
 Distance between 88 and 89 is 16.37524186080926
 Distance between 89 and 90 is 73.55828408248796
 Distance between 90 and 91 is 93.52166880974697
 Distance between 91 and 92 is 16.39136434223826
 Distance between 92 and 93 is 70.81952151772842
 Distance between 93 and 94 is 85.93465353371711
 Distance between 94 and 95 is 46.135525300575054
 Distance between 95 and 96 is 35.126238882066495
 Distance between 96 and 97 is 83.55741906617268
 Distance between 97 and 98 is 26.599960093955016
 Distance between 98 and 99 is 37.30956820977698
 Distance between 99 and 100 is 3.7189036018697736
 Distance between 100 and 1 is 65.59379966124847
 Distance between 1 and 2 is 68.56764645953658
 Distance between 2 and 3 is 64.86986854464868
 Distance between 3 and 4 is 35.3447589608417
 Distance between 4 and 5 is 40.60077639651735
 Distance between 5 and 6 is 73.24243727512076
 Distance between 6 and 7 is 34.48250914594238
 Time is 0.0030012130737304688
 Total distance is: 5084.461369051878

For greedy algorithm

Time is 0.004000186920166016

Shortest path: [7, 63, 76, 15, 46, 70, 75, 74, 1, 86, 33, 48, 90, 52, 40, 84, 13, 50, 47, 5, 21, 12, 66, 35, 49, 29, 96, 71, 36, 65, 19, 27, 23, 59, 20, 95, 55, 4, 56, 31, 87, 72, 57, 82, 34, 3, 43, 64, 62, 45, 51, 97, 93, 6, 77, 88, 85, 25, 26, 22, 89, 79, 9, 10, 83, 100, 73, 99, 53, 58, 67, 32, 14, 54, 91, 18, 41, 61, 24, 16, 92, 78, 2, 30, 37, 42, 38, 17, 11, 80, 28, 94, 44, 81, 69, 39, 8, 98, 68, 60, 7]

Total distance: 982.958671338802

For two_opt algorithm

Time is 2.408505439758301

Optimized path: [7, 62, 45, 51, 93, 97, 6, 88, 77, 85, 25, 26, 22, 68, 53, 89, 79, 98, 64, 43, 3, 82, 34, 57, 72, 87, 31, 56, 4, 8, 9, 10, 83, 99, 73, 100, 30, 2, 37, 58, 67, 32, 14, 54, 18, 91, 41, 61, 24, 16, 92, 78, 42, 69, 81, 44, 94, 28, 80, 11, 17, 38, 39, 55, 95, 20, 59, 29, 96, 71, 36, 65, 19, 35, 49, 66, 12, 21, 5, 23, 27, 50, 13, 47, 84, 40, 52, 90, 48, 33, 86, 1, 46, 70, 75, 74, 15, 76, 63, 60, 7]

Total distance: 842.0855160404456

3)

Dla two-opt pogorszenie czasu wynosi 803000%

Dla two-opt poprawa odległości wynosi 604%

Dla algorytmu zachłannego pogorszenie czasu wynosi 30%

Dla algorytmu zachłannego poprawa odległości wynosi 517%

Najlepszym algorytmem okazał się algorytm two-opt, dla niego ścieżka okazała się najkrótsza. Natomiast najszybszym algorytmem jest przejście w kolejności z pliku.

Przejście zgodnie z kolejnością nie wymaga, żadnych dodatkowych działań podczas planowania dlatego jest on najszybszy, lecz nie zyskuje on nic pod kątem długości ścieżki, dlatego całkowita długość jest zdecydowanie większa niż zastosowanie jakiegoś algorytmu, który to zoptymalizuje. Na przykład algorytmu zachłannego lub two-opt.

Najlepszym algorytmem jest algorytm two-opt, ponieważ analizuje on całą dostępną trasę prowadzącą przez każde miasto i wybiera tą, która jest najbardziej optymalna, przez co trasa jest najkrótsza, lecz czas potrzebny na przeanalizowanie możliwości znacznie wydłuża czas

działania programu.

Według mnie w tym przypadku najbardziej optymalne jest zastosowanie algorytmu zachłannego, ponieważ całkowita długość trasy jest tylko o 100 jednostek większa od algorytmu two-opt, a czas wykonania algorytmu jest porównywalny z przejściem przez miasta w kolejności z pliku.

Na tym laboratorium nauczyłem się jak wykorzystywać algorytmy optymalizujące długość ścieżki podczas przejścia przez kilka punktów. Dowiedziałem się również, że jest to bardzo ważne, ponieważ niewielkim kosztem czasu, można znacznie skrócić jakąś trasę.